

ROLL NO.:

NAME:

Return PART-A in 60 min.



Birla Institute of Technology & Science, Pilani Hyderabad Campus
Computer Science and Information Systems Department
COMPREHENSIVE EXAM – 2023-24

CS F407 Artificial Intelligence

PART - A. SET-A

Correct answer: +1 Wrong answer: $-\frac{1}{2}$

REPRESENTATION*	Rules of quantification		Match
1. $\forall x (P(x) \rightarrow Q(x))$	A. Universal instantiation	1	C
2. $\exists x P(x) \wedge \exists y Q(y)$	B. Existential instantiation	2	E
3. $\neg \forall x \neg P(x)$	C. Universal generalization	3	D
4. $(\forall x P(x)) \rightarrow Q(a)$	D. Negation of universal quantification	4	A
5. $\exists x (P(x) \wedge Q(x))$	E. Existential generalization	5	B

REPRESENTATION*	FACTS		Match
6. $\forall x (P(x) \rightarrow Q(x))$	A. If everyone is Sincere, then John is sincere.	6	D
7. $\exists x P(x) \wedge \exists y Q(y)$	B. There is a student who is both diligent and intelligent.	7	E
8. $\neg \forall x \neg P(x)$	C. It is not the case that everyone is unhappy.	8	C
9. $(\forall x P(x)) \rightarrow Q(j)$	D. Every person likes Music.	9	A
10. $\exists x (P(x) \wedge Q(x))$	E. There exists a person who likes Music, and there exists someone who enjoys Travelling.	10	B

*Note: Match relevant quantifiers; Ignore relevance of symbols/variables P, Q ...

REPRESENTATION	FACTS		Match
11. $P \wedge (Q \vee R)$	A. If I study (P) and understand the material (Q), then I will succeed in the exam (R).	11	B
12. $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$	B. Either it is raining (P), and I have an umbrella (Q), or I am indoors reading a book (R).	12	E
13. $(P \rightarrow Q) \wedge (\neg Q \rightarrow \neg P)$	C. Either I will go to the party (P), or if I don't go, I'll stay home (Q) and watch a movie (R).	13	D
14. $(P \vee Q) \wedge (\neg P \vee R)$	D. If it is sunny (P), then I am happy (Q), and if I am not happy ($\neg Q$), then it is not sunny ($\neg P$).	14	C
15. $(P \wedge Q) \rightarrow R$	E. The statement "It is not the case that both A (P) and B (Q)" is equivalent to "Either A is false or B is false."	15	A

PTO

21	Write the pseudocode of EM algorithm; clearly define the symbols and notations used. [5marks]  EM Algorithm Expectation Step $p(j x_i, \Theta) = \frac{\alpha_j f_j(x_i \theta_j)}{\sum_{k=1}^K \alpha_k f_k(x_i \theta_k)}$ $f_j(x \theta_j) = \frac{1}{(2\pi)^{d/2} \Sigma_j ^{1/2}} e^{-\frac{1}{2}(x - \mu_j)^T \Sigma_j^{-1} (x - \mu_j)}$ Maximization Step $\mu_j^{new} = \frac{\sum_{i=1}^N x_i p(j x_i, \Theta^{old})}{\sum_{i=1}^N p(j x_i, \Theta^{old})}$ $\Sigma_j^{new} = \frac{\sum_{i=1}^N p(j x_i, \Theta^{old}) (x_i - \mu_j^{new})(x_i - \mu_j^{new})^T}{\sum_{i=1}^N p(j x_i, \Theta^{old})}$ $\alpha_j^{new} = \frac{1}{N} \sum_{i=1}^N p(j x_i, \Theta^{old})$ BITS Pilani, Hyderabad Campus Only pseudocode with partial symbols.. [3 marks] 1. Initialize estimates for $\theta := \pi, \mu_1, \sigma_1, \mu_2, \sigma_2$ 2. (Expectation) Compute the responsibilities for each data point $\gamma_i = \frac{\pi \phi(x_i; \mu_2, \sigma_2)}{(1 - \pi) \phi(x_i; \mu_1, \sigma_1) + \pi \phi(x_i; \mu_2, \sigma_2)}$ 3. (Maximization) Update the estimates for the parameters using the maximum-likelihood estimator formula. All sums are taken across the data indexed by i and are just means/standard deviations weighted by the responsibilities γ $\mu_2 = \frac{\sum \gamma_i x_i}{\sum \gamma_i} \quad \sigma_2 = \frac{\sum \gamma_i (x_i - \mu_2)^2}{\sum \gamma_i} \quad \pi = \frac{1}{n} \sum \gamma_i$ 4. Repeat steps 2 and 3 until the parameters converge to a local optimum
22	Derive the upper bound of Entropy and Gini Index, for multi-class classification with each class having same probability [5 marks] Upper Bound Entropy and Gini Index are 1 and 0.5 respectively when data has the highest uncertainty One thing to note here is that these values for the upper bound will only hold in case of binary classification. In the case of n classes which are equally likely each having probability $(1/n)$, the upper bound will be a function of n, as shown below. • For multi-class classification with each class having same probability, the upper bound of Entropy will be: $\begin{aligned} \text{Entropy} &= -[p_1 * \log_2(p_1) + \dots + p_n * \log_2(p_n)] \\ &= -\left[\left(\frac{1}{n} * \log_2 \frac{1}{n}\right) + \left(\frac{1}{n} * \log_2 \frac{1}{n}\right) + \dots\right] = -\log_2 \frac{1}{n} \\ &= \log_2 n \end{aligned}$ • For multi-class classification with each class having same probability, the upper bound of Gini Index will be: $\begin{aligned} \text{Gini Index} &= 1 - [p_1^2 + p_2^2 + \dots + p_n^2] = 1 - n * \left[\left(\frac{1}{n}\right)^2\right] \\ &= 1 - \frac{1}{n} \end{aligned}$